CSC501 C++ for Engineers
Semester 2 – 2025

Lab Exercise 2

Tutorial Questions

Task 1: Multiple Choice

1. The rules you must follow when using a programming language are called it's ______________________ .

   a. guidelines c. regulations
   b. procedures d. syntax

2. Which of the following declares a variable that can store an integer?

   a. int quantity = 0; c. quantity = 0;
   b. integer quantity = 0; d. Int quantity = 0;

3. A C++ statement must end with a ______________________ .

   a. colon c. period
   b. comma d. semicolon

4. The declaration statement for a named constant requires ______________________ .

   a. a data type c. a value
   b. a name d. all of the above

5. Which of the following creates a variable that can store real numbers?

   a. double totalDue = '0.0'; c. double totalDue = "0.0";
   b. double totalDue = 0.0; d. totalDue = 0.0;

6. Which of the following is a valid name for a variable?

   a. amount-sold c. 1stQtrAmountSold
   b. amountSold d. both b and c

7. Which of the following declares a char named constant called TOP_GRADE?

   a. const char TOP_GRADE = 'A'; c. const char TOP_GRADE;
   b. const char TOP_GRADE = "A"; d. both a and c

8. If you use a real number to initialize an int variable, the real number will be ______________________ before it is stored in the variable.

   a. demoted c. reduced
   b. promoted d. upgraded

Task 2: Short Answer question

1. Indicate if the following identifiers are valid or invalid. If invalid, state the rule which it violates:

   a) Power
   b) density
   c) m1234
   d) 1234 f
   e) total
   f) Absval
   g) b34a
   h) 4ab

2. Determine data types appropriate for the following data:

   a) The average of four grades
   b) The number of days in a month
   c) The length of the Golden Gate Bridge
   d) The numbers in a state lottery
   e) The distance from Brooklyn, NY to Newark, NJ
   f) The single-character prefix that specifies a component type

3. Rewrite each of these declaration statements as three separate declarations:

   a) int month, day = 30, year;
   b) double hours, volt, power = 15.62;
   c) double price, amount, taxes;
   d) char inKey, ch, choice = 'f';

Task 3: Expressions

1. For the following correct algebraic expressions and corresponding incorrect C++ expressions, find the errors and write corrected C++ expressions:

Algebra	C++ Expression
a. (2)(3) + (4)(5)	(2)(3) + (4)(5)
b. $\frac{6 + 18}{2}$	6 + 18 / 2
c. $\frac{4.5}{12.2 - 3.1}$	4.5 / 12.2 - 3.1
d. 4.6(3.0 + 14.9)	4.6 (3.0 + 14.9)
e. (12.1 + 18.9)(15.3 - 3.8)	(12.1 + 18.9) (15.3 - 3.8)

2. Determine the values of the following integer expressions:

   a. `3 + 4 * 6`
   b. `3 * 4 / 6 + 6`
   c. `2 * 3 / 12 * 8 / 4`
   d. `10 * (1 + 7 * 3)`
   e. `20 - 2 / 6 + 3`
   f. `20 - 2 / (6 + 3)`
   g. `(20 - 2) / 6 + 3`
   h. `(20 - 2) / (6 + 3)`
   i. `50 % 20`
   j. `(10 + 3) % 4`

3. Assume that:

   amount stores the integer value 1

   m stores the integer value 50

   n stores the integer value 10

   p stores the integer value 5.

   Evaluate the following expressions:

   a. `n / p + 3`
   b. `m / p + n - 10 * amount`
   c. `m - 3 * n + 4 * amount`
   d. `amount / 5`
   e. `18 / p`
   f. `-p * n`
   g. `-m / 20`
   h. `(m + n) / (p + amount)`
   i. `m + n / p + amount`

4. What is the value of x and y after each of the statements is executed. Assume the values of **x** and **y** is **6** and **10**, respectively, **before each statement is executed**.

   a) x /= y

   b) x %= y

Practical Exercise

Task 1

(General math) The area of an ellipse is given by this formula:

$$\textbf{Area} = \pi\, \mathbf{a}\, \mathbf{b}$$

Using this formula, write a C++ program to calculate the area of an ellipse having a minor axis, **a**, of **2.5 inches** and a major axis, **b**, of **6.4 inches**

Task 2

Write a C++ program to calculate and display the value of the slope of the line connecting two points with the coordinates (3,7) and (8,12). Use the fact that the slope between two points at the coordinates (x1,y1) and (x2,y2) is

slope = (y2 - y1) / (x2 - x1)

Your program should produce the display something similar to the following:

The value of the slope is xxx.xx

Try out your program with different coordinate values as well.

Task 3

A model of worldwide population growth, in billions of people, since 2000 is given by this formula:

Population = 6.0 e0.02[Year - 2000]

Using this formula write, compile, and run a C++ program to estimate the worldwide population in the year 2012. Verify the result your program produces by calculating the answer manually. After verifying that your program is working correctly, use it to estimate the world's population in the year 2023.

Task 4

Write, compile, and run a C++ program that displays the following prompts:

Enter the miles driven:

Enter the gallons of gas used:

After each prompt is displayed, your program should use a cin statement to accept data from the keyboard for the displayed prompt. After the number for gallons of gas used has been entered, your program should calculate and display the miles per gallon (mpg). This value should be included in a message and calculated by using the formula:

miles per gallon = miles / gallons used.

Verify your program by using the following test data:

Test data set 1: miles = 276, gas = 10 gallons

Test data set 2: miles = 200, gas = 15.5 gallons

After finishing your verification, use your program to complete the following chart. *(Make sure to convert the miles driven to kilometers driven and gallons used to liters used, and then compute the kilometers per liter.)*

Miles Driven	Gallons Used	Mpg	Km Driven	Liters Used	Km/L
250	16.00				
275	18.00				
312	19.54				
296	17.39				